

ENCRYPTED TRAFFIC ANALYSIS FOR INTRUSION DETECTION IN IOT DEVICES

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May 2021

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INTRODUCTION

Traditional intrusion detection systems base their operations on signatures, which are, basically, a pattern defined, based upon a specific set of characteristics regarding the packets in a chunk of traffic. Those characteristics are retrieved directly from packet headers or packet payloads, in order to form a signature, relying on the frequency of a repeating pattern.

Their functionality has taken a great hit after the introduction of encryption of packets in modern networks, thus the optimization of such procedures is deemed to be of great importance, especially if we take into consideration the great extent that computer networks play in our lives in the recent years and for years to come.

In this work, I am using a dataset that contains infected traffic of an IoT network. My main goal is to come up with signatures, regarding the payload size sequences of network flows containing packets that make use of the TCP protocol and aim to detect the malicious activity of 2 different Botnets. The purpose of this work is to mark the importance of using unencrypted data which reside inside encrypted packets, in the TCP packet headers, as well as, the ability of using such information to detect malicious activity in a IoT network.

Moving on, in Section 2, I am going to, briefly, describe the dataset that was used for my thesis project and expand more on the process of extracting all the valuable information needed to determine the existing network flows from a chunk of traffic and also create the TCP payload size sequences of such flows. In Section 3, I am presenting the results of analysing all the information gathered, as well as, exhibit all the patterns that got revealed during the analysis phase. Section 4, describes my attempt to evaluate the results that are proposed in Section 3 and its shortcoming. Finally, in Section 5, I am going to discuss the overall conclusions of this work, as well as, possible limitations that arose during this whole process.

THE EXPERIMENT

In this chapter, I am going to provide a description of the experiment. More specifically, I am going to expand on matters such as, the dataset that was used and the process of extracting the information from the packets residing in the pcap files of the dataset.

2.1 DATASET DESCRIPTION

The IoT-23 dataset was created by the Stratosphere Laboratory, AIC group, FEL, CTU University, Czech Republic. The IoT-23 dataset consists of network traffic generated by the interaction of different malware types with a Raspberry Pi device. For my case, I used the traffic that was generated using the Tori and Mirai botnets. In the table below, I am presenting a short overview of the samples that were used in this project.

Table 1: Samples used for signature extraction. The first column contains the names of the samples, as they were provided by the creators of the dataset, the second column contains the malware name of the infected samples, while the third one contains abbreviations of the initial samples' names as I refer to them in this document.

Full Name	Malware Type	Short Name
CTU-IoT-Malware-Capture-34-1	Mirai	Mirai_34
CTU-IoT-Malware-Capture-44-1	Mirai	Mirai_44
CTU-IoT-Malware-Capture-48-1	Mirai	Mirai_48
CTU-IoT-Malware-Capture-49-1	Mirai	Mirai_49
CTU-IoT-Malware-Capture-20-1	Tori	Tori_20
CTU-IoT-Malware-Capture-21-1	Tori	Tori_21
CTU-IoT-Malware-Capture-43-1	Mirai	Mirai_43
CTU-IoT-Malware-Capture-52-1	Mirai	Mirai_52

For the aforementioned malicious traffic scenarios, a pcap file is provided, accompanied by a set of files with additional information regarding those separate traffic scenarios. Out of these files, I mostly used a file named conn.log.labeled. This file was generated by the creators of the dataset using zeek, which is a well-known open source network security monitoring tool. The conn.log.labeled file contains all the network flows residing in each specific traffic chunk. To be more precise, the network flows contained in the zeek file are bidirectional

and there may be multiple instances of the same flows, based on different times intervals. A stripped down of this file is used for this work, which contains only the networks flows that are TCP related.

2.2 PACKET PROCESSING

From a more technical aspect, 8 pcap files were used for this work, two of those containing Tori Botnet's traffic and the rest of them containing Mirai Botnet's traffic. A packet analyzer was implemented, using libpcap, in order to obtain the needed information from the packets. Moving on, I am going to expand more on the functionality of the packet analyzer.

2.2.1 *Packet Filtering*

At this point, it is important to explain the nature of the traffic that I had to handle in order to produce the network flows that will be analyzed later on. First, let us define the main goal, which is to produce the TCP payload size sequences for each network flow that resides in the available pcap files. Unfortunately, the traffic captures contain traffic which is not TCP related, thus such packets got removed. Aside from that, even some TCP packets are filtered out due to the need of keeping the practical view of these TCP communications. To sum this up, I removed any packet that complies with any of the following,

- Non-TCP packets
- Packets with TCP payload size equal to zero
- Retransmitted TCP packets

Non-TCP packets are obviously removed, because they were not the main focus. Other than that, by removing the zero size TCP packets we ensure that we filter out acknowledgement packets as well as simplifying the payload size sequences that will be generated, by not including either, single zero-length size packets or sequences of such payloads in the corresponding entries. To add to that, TCP retransmissions are also excluded in attempt to keep the generated network flows unaffected by the unpredictable behaviour of an IoT network.

2.2.2 Network Flow Generation

After discarding the packets, as described above, all the network flows are identified based on the following characteristics,

- Source ip
- Destination ip
- Source port
- Destination port
- Protocol

Moving on, the TCP payload size sequences are generated by assigning each TCP payload size of every processed packet, as well as the relative timestamp of the packet, on the corresponding network flow that was already identified. Finally, the packet analyzer creates a spreadsheet containing all the information gathered for each network flow from a specific pcap file. The entries of the spreadsheet contain some additional information for each one of the flows,

- Flow Id
- Infected device's name (This is always a Raspberry Pi)
- Total number of packets in flow
- Total number of bytes in flow
- and the TCP payload sequence for this flow

2.3 FLOW LABELING

The final step, before analysing the results, was to label the identified flows of each pcap into malicious and benign. In order to do so, a simple python script was implemented, which received the spreadsheet, that was previously created by the packet analyzer, and the corresponding zeek report (conn.log) of each pcap file. The script had 2 main goals:

- Label the flows of the spreadsheet, that received as an input, as either malicious or benign, as well as, copy the label that describes the nature of the attack, which was also included in the conn.log.labeled file.
- Split the benign flows from the malicious ones and write both of them in separate csv files in order to proceed with analysis

Before moving on, I would like to point out the fact that some of the flows had multiple instances of them residing in the `conn.log.labeled` file. Some of them were labeled as both malicious and benign. I completely discarded those entries in an attempt to keep the results of the forthcoming analysis results as much as apparent as possible.

ANALYSIS

In this Section, I am presenting my findings regarding the two different malware families, Mirai and Tori Botnets. For each malware family, I will describe communication patterns that took shape in the spreadsheet containing the network flows and based on these patterns I will propose signatures that aim to the detection of the such malware in an IoT network.

3.1 TORI INFECTED TRAFFIC

For this malware family, a total of 14, non-bidirectional malicious tcp flows were generated. It is important to mention that these flows are not bidirectional. Out of these 14 flows, 11 of them are labeled as malicious, while the rest of them are benign. The findings, which are presented below, arised from the Tori_20 and Tori_21 samples.

3.1.1 *Total number of Packets/Bytes sent for pair of malicious flows*

To begin with, I noticed that a correlation exists between some flows, regarding the total number of packets/bytes sent. To be more precise, browsing through the entries of the spreadsheet containing the malicious flows, the fields of total number of packets and bytes sent for pairs of malicious flows with the same ips and ports values but the opposite direction, seem to hardly differ from each other. In this hypothesis I will examine such pairs of flows that have the same ips and ports but the opposite direction. An instance of such a pair is displayed below,

Table 2: A pair of network flows with the same ip and port values which represent each direction of the whole communication between these ips over these ports.

src ip	dest ip	src port	dst port	total packets	total bytes
46.101.251.172	192.168.1.199	80	59238	84	121390
192.168.1.199	46.101.251.172	59238	80	1	148

For the majority of the malicious pairs of flows, the total number of bytes sent is equal. This is not the issue for 2 pairs of malicious flows. In the first case, the total bytes values of the flows within the pair differ. In the second case, there is a malicious flow without a pair in order to compare their total bytes value. Overall, the total bytes values match in 4 out of 6 (66,66%) malicious flow pairs.

Now, for almost every pair of flows, I notice that the number of total packets being sent between those ips using the same ports are equal or differ by 1. This is the issue for 5/6 (83,3%) malicious flow pairs, including the malicious flow that does not have a pair.

At this point, is important to mention, that the benign flows found by the packet analyzer do not comply with the above observations which is a way to distinct malicious flows from benign flows, but it is not wise to believe that this assumption is 100% concrete due to the fact that the number of benign flows were only 3 in total (from both samples analyzed by the packet analyzer).

Finally, the total packets sent and total bytes sent values, in a pair of flows, provide a way to detect the existence of Tori Botnet taking advantage of infected devices. Unfortunately, this hypothesis need to be furtherly tested, using a greater number of both malicious and benign flows, in order to obtain more reliable results that back up my hypothesis.

Moreover, the above heuristic does not provide us with a real-time detection method. We can only detect a Tori Botnet's activity only after the communication has ceased, for a pair of flows, due to the way that we generate those total values.

3.1.2 *Payload Sequences Analysis*

In this section, I examine each malicious flow as it is, not as a pair. At a first glance, the benign flows can be picked apart from the malicious ones only by looking at the TCP payload sequence of each flow. More specifically, the benign flows contain a small number TCP payloads, while also the payload size values are different than those seen in the malicious flows.

In contrast to the benign flows, all malicious flows contain a much greater number of TCP payloads. What is more, the malicious flows' payload sequences seem to form a specific payload size pattern, where the initial 2 or 3 payloads have certain lengths, within a specific range of values, and the rest of them are equal to 37 bytes. Those 2 or 3 initial payload sizes in each payload sequence, that are not equal to 37 bytes long, are equal to 1460 bytes length in 8/11 (72,72%) malicious flows. To sum up the above, we can assume that a network flow is malicious, when the following attributes are met in this particular network flow,

1. The first TCP payload size in the payload sequence is 1460 byte long.

2. The payload sequence of this particular flow, mainly consists of many consecutive packets with TCP payload size of 37 bytes, except from the first 2 or 3 payload sizes.

Thus I am proposing a signature such as,

[1460, w/e, 37, ..., 37]

OR

[1460, w/e, w/e, 37, ..., 37]

where "1460" is the payload size of the first TCP payload size in this flow, followed by one or two payload sizes of either the same payload size or close to that of the first payload, then followed by a long consecutive sequence of 37-bytes-long TCP payload sizes.

The use of this signature is possible to detect the existence of a malicious flow in real-time, only by tracking the TCP payload size of each packet in a specific flow.

The results which were provided for this heuristic are more solid than those of section 3.1.1, because the conclusions were drawn by a bit larger pool of samples (individual flows not pairs of flows) but still a greater number of flows, would provide a greater degree of trust in my hypothesis.

To sum this up, based on these assumptions, we are able to discern a malicious flow from a benign one, either by checking both the total packets and total bytes exchanged in both directions of a network flow (same ips, same ports, different directions), or by examining the TCP payload size of a specific flow in correlation with the signature that was proposed.

3.2 MIRAI INFECTED TRAFFIC

Overall, 4 different pcap files were analysed for signature extraction. Each pcap file seems to hold a unique traffic scheme when an infected device interacts with an attacking device that belongs in a Mirai Botnet.

For this malware family, I followed a slightly different process of analysing the flows derived from the pcap files. While maintaining the focus on pairs of flows with the same ips and ports, I did not put my main focus on the total packets and total bytes values for the Mirai Botnet's generated flows. Instead, the main effort was put into describing other patterns that occurred inside the payload sequences of such pairs and other characteristics that may correlate with the existence of such payload sequences. The main reason, of changing my approach was that, due to the greater number of flows I had at my disposal, there were much more traffic schemes to analyse.

The generated flows were 6125 in total, where 6078 of them were labeled as malicious and 47 as benign. At this point is important to mention that each pcap file had an almost unique traffic scheme. Meaning that the majority of the patterns that took shape in the payload sequences of the flows of a specific pcap file, will not appear in the payload sequences of the flows generated by another pcap which was also used.

Moving on, I am going to describe some general observations between all traffic schemes, some distinctive payload sequences repeating in each pcap and propose some signatures for the detection of Mirai's malicious activity in an IoT network.

3.2.1 General Observations

First of all, the most pcap files that got processed, produced network flows between a very limited number (2 or 3 at the most) of ips, with only one sample, Mirai_48, being an exception of that, due to the fact that the infected device seemed to have established many connections with a lot of different ips.

Other than that, the set of used port numbers of the infected devices seems to be a broad one. Still, the ports used by the infected devices, displayed a certain degree of homogeneity. It is important to note that the port values of the infected devices between 2 different pairs flows had a small offset, which was usually 2 the majority of times.

What is more, the port numbers used by the devices that were a part of Mirai's Botnet used much less port numbers, 4 different in total. Those were, 6667, 80, 4554 and 23. The use of those port numbers makes me able to discern what kind of upper layer protocols were used by Mirai in order to perform its malicious activity. Namely, those were Irc, Http, Msfrs and Telnet. Irc and Telnet protocols have formerly used with malicious intent.

Unfortunately, the number of benign flows for this malware family were also not great enough in order to be able to confidently compare their characteristics with the malicious ones, thus it would not be wise to put too much effort into comparing malicious and benign flows. What I could point out though, would be the fact that the TCP payload sizes met in the benign flows of each pcap had completely different payload values in them, in comparison to the malicious ones. Still, this was not the issue for all the pcaps, there was only one exception that I am going to refer to later on, on Section 3.2.2.3.

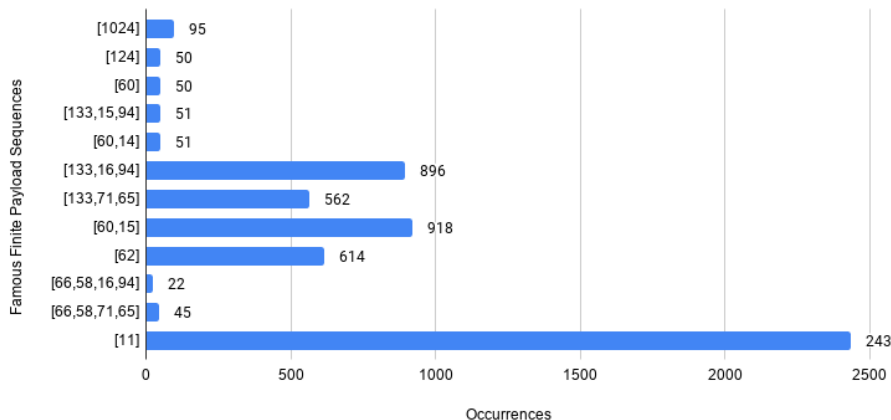
3.2.2 Payload Sequence Analysis

First of all, from each sample the most common payload sequences were detected from each flow that was available. Those, were either payload sequences that had exactly the same payload sequences in terms of number of TCP payloads and TCP payload sizes or payload sequences that had different number of TCP payloads but the sizes of those were of a specific length, thus leading to them being identifiable among others that did not show any type of consistency regarding their sizes or other characteristics.

3.2.2.1 "Famous" finite Payload Sequences

In this section, I am displaying payload sequences with a finite number of payload sizes, which, most of them, were extracted by Mirai_34 sample. In Table 3 and Figure 1, the famous finite payload sequences are displayed as well as the times that they were encountered in the aforementioned sample.

Figure 1: Famous finite payload sequences and their occurrences



Now, I would like to specify a few things for the famous finite payload sequences with ids 1 and 12.

Table 3: Famous finite payload sequences

Id	Sample	Famous Finite Payload Sequences	Malicious Activity
1	Mirai_34	[1024]	DDoS
2	Mirai_34	[124]	C&C
3	Mirai_34	[60]	C&C
4	Mirai_34	[133, 15, 94]	C&C
5	Mirai_34	[60, 14]	C&C
6	Mirai_34	[133, 16, 94]	C&C
7	Mirai_34	[133, 71, 65]	C&C
8	Mirai_34	[60, 15]	C&C
9	Mirai_34	[62]	C&C
10	Mirai_34	[66, 58, 16, 94]	C&C
11	Mirai_34	[66, 58, 71, 65]	C&C
12	Mirai_49	[11]	C&C

- The first sequence with id 1, is labeled as a DDos attack where all the packets are sent from the infected device (Raspberry Pi) to the attacking device using the http protocol, this means that we have only single-direction instances of such flows. I assume that this attack's nature is to keep the infected device occupied from doing what it was supposed to do at the given time. Such flows are sequential to one another inside the spreadsheet of the collected data.
- The flows having a payload sequence which is the same as the one with id 12, are the only ones that did not come from the same sample, unlike all the others. These flows were also sequentially existed in the according spreadsheet and they were bidirectional. Their malicious activity was labeled as C&C and the only port used by the attacking device is always 4554, consequently the msfrs protocols is used in the upper layers of such packets.

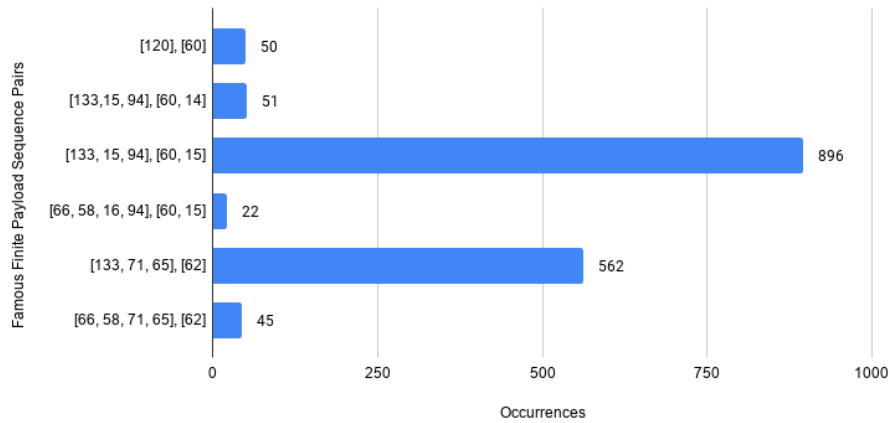
3.2.2.2 Pairs of "Famous" Payload Sequences

Furthermore, many of the above sequences can be paired with a sequence, which is also present in Table 3, where the paired sequence is the one of the opposite direction, meaning that this flows share the same ips & ports. Many of those pairs are obvious if anyone takes notice on the counters of the occurrences, in Figure 1. For example, if we take a closer look at the lines 4 and 5 we can tell that this is a pair of such flows. Below, all those pairs are displayed.

Table 4: Pairs of malicious payload sequences and their occurrences. The first payload sequence in each pair represents the packets sent from the attacking system and the second sequence is the packets that the infected device sends to the attacking system.

Pair ID	Famous Finite Payload Sequence Pairs
1	[120], [60]
2	[133, 15, 94], [60, 14]
3	[133, 16, 94], [60, 15]
4	[66, 58, 16, 94], [60, 15]
5	[133, 71, 65], [62]
6	[66, 58, 71, 65], [62]

Figure 2: Pairs of famous finite payload sequences and their occurrences



Now, I would like to point out a few things for the pairs of the famous payload sequences,

1. First of all, all these sequences came from the same sample, Mirai_34.
2. In all of these flows the attacking device uses the 6667 port in order to send or receive the packets, thus I come to the conclusion that the Irc protocol is used by the attacking device.
3. For some of those pairs, a single instance of both directions of another pair is encountered, which is a repeating event. For example, in Figure 2, we can observe that we come across multiple instances of flows that look like the pair of flows with id 5 and after 10 instances of those, the next to 2 flows have payload sequences equal to those of pair 6. To conclude, this is also the issue for another pair of flows, those with id 3 and 4.

4. It is also obvious that for these pairs, the total packet & bytes values do match.
5. All these pairs of flows have their malicious activity been labeled as C&C as reported in corresponding Zeek file of this sample.

Finally, the payload sequences displayed in the Tables 3 and 4 can be used as signatures by intrusion detection systems. Having said that in order to make proper use of these signatures, the designer of such systems should also consider how to integrate those, meaning that the system could either just scan the TCP payloads of incoming packets of a specific flow with a specific direction or it could make use of other characteristics of these flows, such as the ports that the packets are received or sent from and the corresponding, opposite direction, flows of an already suspicious flow, in an attempt to make a much more trustworthy detection.

Figure 3: **The pair of flows with id 4 shows up once in a while, in between the pairs of flows with id 3**

133	71	65	
62			
66	58	71	65
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
133	71	65	
62			
66	58	71	65
62			

3.2.2.3 Common Pairs of Payload Sequences

As I mentioned before, each sample that was used to generate the flows seemed to have captured a specific kind of activity of Mirai's Botnet making the sequences derived from them being somewhat unique when comparing them with sequences of other samples. Despite that, some payload sequences with a specific pattern seem to be present in

every sample. To be more precise, this was the issue for a specific pair of flows.

The directions of these pairs for the transmitted packets from the attacking device towards the infected device had, mainly, TCP payload values of 1448 bytes and only a few of them were 2896 bytes long which is 1448 doubled, but this doubles were extremely occasional. What is more, the last TCP payload values for each of these sequences were always a value less than 1448 which seemed to be random for each of these flows. The total packets values of these flows fluctuated between 52 and 145 packets per such flow, while the total bytes values took values from 74766 to 209560 bytes. The opposite direction flows of these pairs always had 1 TCP payload value, ranging from 147 to 152 bytes.

The port that the attacking device was using to receive or transmit those packets was port 80, thus we assume that Mirai is making use of http to transmit such malicious TCP packets.

Moreover, the number of such pairs in 3/4 pcap is 11 and 12 for the remaining one. At this point, it is important to underline the fact that these 12 pairs in the only pcap that has more than 11 of those, are found to be labeled as benign by the corresponding zeek report which leaves us with the impression that Mirai is possibly trying to hide some of its malicious activities despite, maybe, using one of its malicious procedures to transmit non-malicious packets instead of malicious ones. Furthermore, all the malicious instances of these pairs have their malicious activity labeled as File-Download or C&C.

Finally, based on what is already mentioned above, the proposed signature would be,

$$[1448, 1448, \dots, 1448, x], [\sim 150]$$

OR

$$[2896, 1448, \dots, 1448, x], [\sim 150]$$

where the x is a TCP payload size which is always less the 1448 bytes that does not show any consistency and the opposite direction flow contains only one TCP payload packet with a payload size value around 150 bytes. I would like to note that, the 2896 byte value might appear in the first or in any other part of the sequence, but as I already mentioned, these TCP payload size value is extremely rare in such sequences and there is a high chance of not even existing. What is more, the number of total packet or bytes transmitted for such flows is always different. For this specific signature, we should always keep in mind that they might not reveal a flow that contains malicious packets, but due to the fact that it is encountered inside different traffic scenarios of malicious traffic it could be a great indicator that a device has been compromised by a Mirai Botnet even if the flow itself does not transmit any malicious packets at a given time.

3.2.2.4 *Pattern Recognition*

Now, I would like to expand on the payload sequences of a specific sample, the Mirai_48 sample. The reason why I am, specifically, expanding on this certain sample is because it had the most non-consistent payload sequences which were also presenting some kind of repeating behaviour. For this section, I am not referring to some of the flows that Mirai_48 sample also included, because a part of these sample's traffic was the aforementioned payload sequences in Section 3.2.2.3

At this point, it is important to note that the spreadsheet containing the flows of this sample was the one that had the greatest number of unique source and destination ip addresses. This played a pivotal role in the way that the payload sequences changed their values, of payload sizes, and the patterns of those. Consequently, this phenomenon has two possible explanations. Either the attacking device manually changes its ip address before establishing a new flow of communication with the infected device or that the infected device is communicating with lots of different compromised machines. Either way, the main observation still remains, when one of the ip addresses changes this means that the payload sequence's values also change.

On the more analytical aspect of things, different TCP payload size patterns were forming between the infected device and each of the unique ip addresses. The port used by the attacking device in this sample was port 23, thus we have a high usage of Telnet protocol which is also related to the overall malicious activity which resides in this pcap file. The labels, which were provided for these flows by the zeek report were mostly Horizontal Port Scans and the Attack label. A detailed description of all the labels mentioned in this document, is provided at Appendix. All the descriptions are provided by the creators of the dataset.

While observing the spreadsheet that lead me to see that small fractions of similar payload sequences are shaping. In order to dissect the major identical features of these fractions we should take a look at table 5.

Table 5: Fraction of pairs of malicious flows where for a unique ip of an attacking device. The first flow represents the direction: "infected device -> attacking device" while the second one represents the opposite one.

Id	Total Packets	Total Bytes															
1	14	27	1	2	1	2	1	2	1	2	1	2	4	2	4	2	
	6	95	15	24	4	12	2	38									
2	14	28	1	2	1	2	1	2	1	2	1	2	4	2	5	2	
	6	95	15	24	4	12	2	38									
3	14	27	1	2	1	2	1	2	1	2	1	2	4	2	4	2	
	6	95	15	24	4	12	2	38									
4	14	28	1	2	1	2	1	2	1	2	1	2	4	2	5	2	
	6	95	15	24	4	12	2	38									
5	14	28	1	2	1	2	1	2	1	2	1	2	4	2	5	2	
	7	133	15	24	4	12	2	38	38								
6	14	31	1	2	1	2	1	2	1	2	1	2	4	2	8		
	6	133	15	24	16	2	38	38									
7	14	27	1	2	1	2	1	2	1	2	1	2	4	2	4	2	
	6	133	15	24	16	2	38	38									
8	14	28	1	2	1	2	1	2	1	2	1	2	4	2	5	2	
	7	133	15	24	4	12	2	38	38								
9	14	29	1	2	1	2	1	2	1	2	1	2	4	2	6	2	
	6	95	15	24	4	12	2	38									

At a first glance, there is a high resemblance amongst these flows' payload sequences as well as the total packets & bytes fields. For this kind of payload sequences we could easily propose some signatures or rules, but the problem with those is that there is a high possibility for them to be working for payload sequences of a specific ip address of an attacking device. At this point, it would be crucial to observe another fraction and pinpoint the similarities of both those fractions. The table below, display another fraction of Mirai_48 sample.

Table 6: Fraction of pairs just like the one in Table 5.

Id	Total Packets	Total Bytes															
1	12	24	1	2	1	2	1	2	1	2	4	2	4	2			
	5	91	12	22	16	2	39										
2	12	24	1	2	1	2	1	2	1	2	4	2	4	2			
	5	91	12	22	16	2	39										
3	12	25	1	2	1	2	1	2	1	2	4	2	5	2			
	5	91	12	22	16	2	39										
4	12	24	1	2	1	2	1	2	1	2	4	2	4	2			
	6	91	12	22	4	12	2	39									

The main similarity in these two fractions is the fact that, the direction which represents the packets sent by the infected device towards

the ip of device that is a part of Mirai maintains a similar payload sequence. Thus we can define a signature as this one,

$$[1, 2, 1, 2, 1, 2, 1, 2] (1)$$

What is more, the first few packets of the opposite direction flow seem to remain the same for a short period. The problem with this assumption is that it might be consistent enough in order to be observed but it is not consistent enough in order to be considered as a permanent phenomenon.

At last, taking everything that was mentioned, in this subsection, into account, I am certain that in order to be highly confident to declare such traffic as malicious, the need of creating a procedure arises. This procedure should take into account every piece of data that is displayed in the table and is also described in this subsection in general. In addition, this procedure has to process the whole traffic before it comes up with the conclusion that it is malicious, meaning that there is no way of detecting the malicious activity in real time.

The procedure performs the following steps.

1. If a payload sequence arises that resembles the (1) signature check the tcp payload of the corresponding direction.
2. Repeat step 1 for two pairs of flows, that each of them has the same ip addresses but has different ports for the infected device.
3. If the corresponding flows of both pairs have similar payload sequences then we assume that there is a risk of malicious activity being performed
4. Repeat this process until the corresponding flows' of the tcp payload sequences change, while the first three steps produce more and more pairs that do resemble each other, the level of assurance, that the traffic in the pcap is malicious, is growing.

We should also keep in mind the following comments regarding this method,

- The tcp payload sequence of the flow that complies with (1) should remain, more or less, the same for every processed pair of flows, meaning that in each pair, we should have at least one direction's tcp payload sequence looking like (1).
- The total packets and total bytes fields are the ones that show the most consistency in terms of their values during the duration of a fraction, making them appropriate for identifying and matching specific flows that their payload sequence is vaguely altered by Mirai, possibly to avoid detection.

- This method can be applied in real-time but the level of authenticity of this method is increasing as the number of examined flows increases too.

EVALUATION

The main purpose of this Section is to evaluate the signatures that were proposed in Section 3. Unfortunately, the samples of Tori infected traffic were only 2 in total, thus I did not have the opportunity to evaluate the proposed signatures for the Tori Botnet. Other than that, I used the 2 remaining samples of Mirai for this task. Those were the Mirai_43 and Mirai_52 samples. Despite that, based on the fact that each sample seemed to contain a unique traffic scheme the results that I obtained are not the most desirable, due to the fact that the only sequences that are actually present in these 2 samples were the ones described in Section 3.2.2.3. At this point, is important to remind that this kind of sequences were present in all samples used for this work and they have also been labeled both as malicious, in for 7 out 8 samples, and benign. Having said that, the signature of Section 3.2.2.3 has matched with 14 out 45 malicious payload sequences of Mirai_43 sample and with 11 out 43 malicious payload sequences of Mirai_52 sample. I would also like to underline that, the fact that there are malicious payload sequences that do not match with the signature, is not an issue because the main goal of this work is to come up with ways to detect the malicious activity of such Botnet and not to detect all the existing malicious network flows within it.

CONCLUSIONS

In this thesis, the main focus was put on generating network flows, derived from packets of IoT network traffic, in order to be analysed and finally, come up with signatures and a procedure to detect the malicious activity of 2 different kinds of well-known malware. Having said that, the lack of a greater number of benign flows did not provide me with the chance of comparing the malicious network flows with the benign ones, in order to raise the level of trust of the results, to a certain degree. This would have provided a greater degree of assurance for the proposed signatures. Adding to the already-mentioned, this work lacks in the evaluation part due to the limited number of pcap files and traffic schemes, for the selected malware families.

APPENDIX - MALICIOUS LABELS

Below, the malicious labels' description is displayed, as it was provided by the creators of the dataset.

Attack: this label indicates that there was some type of attack from the infected device to another host. Here we are labeling as attack to any flow that, by analysing its payload and behaviour, tries to take advantage of some vulnerable service. For example, a brute force to some telnet login, a command injection in the header of a GET request, etc.

Benign: this label indicates that no suspicious por malicious activities where found in the connections.

C&C: this label indicates that the infected device was connected to a CC server. This activity was detected in the analysis of the network malware capture because the connections to the suspicious server are periodic or our infected device is downloading some binaries from it or some IRC like or decoded orders are coming and going from it.

DDoS: this label indicates that a Distributed Denial of Service attack is being executed by the infected device. These traffic flows are detected as part of a DDoS attack because of the amount of flows directed to the same IP address.

FileDownload: this label indicates that a file is being downloaded to our infected device. This is detected by filtering connections with response bytes more than 3KB or 5KB, normally this is combined with some known suspicious destination port or destination IP known to be a C&C server.

Mirai: this label indicates that the connections have characteristics of a Mirai botnet. This label is added when the flows has similar patterns as the most common known Mirai attacks.

PartOfAHorizontalPortScan: this label indicates that the connections are used to do a horizontal port scan to gather information to perform further attacks. To put these labels we rely on the pattern in which the connections shared the same port, a similar number of transmitted bytes and multiple different destination IPs.

Torii: this label indicates that the connections have characteristics of a Torii botnet. This labeling decision was made with the same parameters as with Mirai but with the difference that this botnet family is less common.